

A Packing of 13 Four-Spaces in $\mathbb{R}^{10}$

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Abstract

We give 13 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.599696. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.584278, obtained by deleting 1 of the 14 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 13 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.599696 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 13$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.584278, obtained by deleting 1 of the 14 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 13/12 = 2.600000$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N13.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 62 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
129984810002390181358599221571341661143682127103294508325643988860579082
754385093947489718294125888966493014243450599635106836591852692737748640
198720923137995014771099052261203872078579337933
```

denominator:

```
500000000000821286565023286834111216327048317601798772310484998403151720
603106977682715908803788467327648670887673239114427208439012117485621314
675439249876763651435546339887195575085081673043
```

B Coordinates

13 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N13.txt`.

```
655753667673 -209665447973 -200431458597 -376181354299483002066 -404836918077 167519583713 264858806924255332058 -136923230391 -454754457220 -587096943124
121250158995 244112519603 -296027238337 -167541460421452911345 -360735196421 -321443110801 3162368736633841092525 90756932870 149460993816 198761063020
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